Assignment No 3

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Q1 Double Number Pyramid

for n = 4

```
        1
      1   2
    1   2   3
  1   2   3   4
    1   2   3
      1   2
        1
```

```
int n = 4
for (int i = 1 ; i ≤ n ; i++) {
    int count = 1
    for (int j = 1 ; j ≤ n ; j++) {
        if (i + j ≥ n + 1) {
            sop (count + " ");
            count++;
        } else {
            sop (" ");
        }
    }
    sop ();
}
for (int i = n - 1 ; i >= 1 ; i--) {
    int count = 1;
    for (int j = 1 ; j ≤ n ; j++) {
        if (i + j ≥ n + 1) {
            sop (count + " ");
            count++;
        } else {
            sop (" ");
        }
    }
    sop ();
}
```

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half upward pyramid

~~loop~~	~~condn~~	~~loop~~	~~condn~~	~~condn~~	~~incre~~
~~i~~	~~i≤n~~	~~j~~	~~j≤n~~	~~if~~	~~decre~~
~~1~~	~~1≤4~~	~~1~~	~~1≤4~~		
			~~True~~		

Row 1 (i = 1)

~~i = 4~~

j	j+i (i=1)	condition	print	incre
1	2	2≥5 ✗	" "	j++
2	3	3≥5 ✗	" "	j++
3	4	4≥5 ✗	" "	j++
4	5	5≥5 ✓	~~1~~ "	j++
~~5~~	~~6~~			

जेव्हा वेळेस j loop मधली cond^n True होत नव्हती count++ होत.

Row 2 (i = 2)

j	j+i (i=2)	condition	print	~~j incre~~
1	2	2≥5 ✗	" "	j++
2	~~3~~ 4	4≥5 ✗	" "	j++
3	~~4~~ 5	5≥5 ✓	1	j++
4	6	6≥5 ✓	2	j++

Row 3 (i = 3)

j	j+i (i=3)	condition	print	incre.
1	4	4≥5 ✗	" "	j++
2	5	5≥5 ✓	1	j++
3	6	6≥5 ✓	2	j++
4	7	7≥5 ✓	3	j++

Row 4 (i = 4)

j	i+j (i=4)	condition	print
1	5	5≥5 ✓	1
2	6	6≥5 ✓	2
3	7	7≥5 ✓	3
4	8	8≥5 ✓	4

→ down word half
i start from n−1, आगे i ही decrement होगा
Row (i = 3)

j	i+j (i=3)	condition	print
1	4	4≥5 ✗	" "
2	5	5≥5 ✓	1
3	6	6≥5 ✓	2
4	7	7≥5 ✓	3

Row (i = 2)

j	i+j	condition	print
1	3	3≥5 ✗	" "
2	4	4≥5 ✗	" "
3	5	5≥5 ✓	1
4	6	6≥5 ✓	2

Row (i = 1)

j	i+j	condition	print
1	2	2≥5 ✗	" "
2	3	3≥5 ✗	" "
3	4	4≥5 ✗	" "
4	5	5≥5 ✓	1

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9. Number Butterfly

n = 4

```
1                   1
1 2               2 1
1 2 3           3 2 1
1 2 3 4 4 3 2 1
1 2 3 4 4 3 2 1
1 2 3       3 2 1
1 2           2 1
1               1
```

```
int n = 4;
for (int i=1 ; i≤n ; i++){
    for (int j=1 ; j≤i ; j++){
        sop(j);
    }
    for (int k=1 ; k≤ 2*(n-i) ; k++){
        sop(" ");
    }
    for (int s=i ; s>=1 ; s--){
        sop(s);
    }
    sop();
}
for (int i=n ; i>=1 ; i--){
    for (int j=1 ; j≤i ; j++){
        sop(j);
    }
    for (int k=1 ; k=2*(n-i) ; k++){
        sop(" ");
    }
    for (int s=i ; s≥1 ; s--){
        sop(s);
    }
    sop()
}
```

Dry run

1st half.
i = 1 to n(4)
1 half outer loop start 1 to n and print n rows.

// 1st triangle loop
j start 1 to i
print(j)

// space loop
k loop start 1 to (2*(n-i))
if i = 1 → 1 to 2*(4-1) = 6 spaces
if i = 2 → 1 to 2*(4-2) = 4 spaces
if i = 3 → 1 to 2*(4-3) = 2 spaces
if i = 4 → 1 to 2*(4-4) = no spac

// 2nd triangle
j start from i to 1
if i = 1 → print 1
i = 2 → print 21
i = 3 → print 321
i = 4 → print 4321

// 2nd half

i start from n-1 to 1

// 1st triange
(j loop)

start from 1 to i

if i = 4 j = 1 to 4 ⟹ 1 2 3 4
i = 3 j = 1 to 3 = 1 2 3
i = 2 j = 1 to 2 = 1 2
i = 1 j = 1 to 1 = 1

// space
k loop start 2 to (2×(n-i))
and space print

2nd triangle
j loop start from i to 1

if i = 4 j = 4 to 1 ⟶ 4 3 2 1
if i = 3 j = 3 to 1 3 2 1
if i = 2 j = 2 to 1 2 1
if i = 1 j = 1 to 1 1

exit loop

Q.3 Cross (X) Number Pattern

for n = 5,

```
1   1
  2 2
    3
  4 4
5   5
```

```
int n=5
int count =1;
for (int i=1 ; i<= n; i++) {
    for (int j=1 ; j<= n ; j++) {
        if (i+j==n+1 || i==j) {
            sop (count);
        } else {
            sop ("  ");
        }
    }
    count ++;
    sop ();
} }
```

Row1 (i=1, count=1)

j	condition	output
1	i==j → 1==1	1
2	false	" "
3	false	" "
4	false	" "
5	i+j=6 → 1+5	1

Row 2 (i= 2 , count = 2)

j	condition	output
1	false	" "
2	i==j → 2==2	2
3	false	" "
4	i+j=6 (2+4)	2
5	false	" "

Row 3 (i = 3, count = 3)

j	condition	output
1	false	" "
2	false	" "
3	i == j (3 == 3)	3
4	false	" "
5	false	" "

Row 4 (i = 4, count = 4)

j	condition	output
1	false	" "
2	i+j = 6 (4+2)	4
3	false	" "
4	i == j (4 == 4)	4
5	false.	" "

Row 5 (i = 5, count = 5)

j	condition	output
1	i+j = 6 (5+1)	5
2	false	" "
3	false	" "
4	false	" "
5	i == j (5 == 5)	5

final o/p

```
1     1
  2 2
    3
  4 4
5     5
```

Q.4 Spiral layer pattern

$n = 3$

3	3	3	3	3
3	2	2	2	3
3	2	1	2	3
3	2	2	2	3
3	3	3	3	3

```
int n = 3;
int row = 2*n-1
int col = row;

for (int i=1; i<=row; i++) {
    for (int j=1; <= col; j++) {

        int top = i-1;
        int bottom = 2*n-1-i;
        int left = j-1;
        int right = 2*n-1-j;

        int s1 = Math.min(top, bottom);
        int s2 = Math.min(right, left);

        int smallest = Math.min(s1, s2);

        sop(n- smallest)
    }
    sop();
}
}
```

Row 1 (i = 1)

j	top	bottom	left	right	smalest	point
1	0	4	0	4	0	3
2	0	4	1	3	0	3
3	0	4	2	2	0	3
4	0	4	3	1	0	3
5	0	4	4	0	0	3

Row 2 (i = 2)

j	smallest	point
1	0	3
2	1	2
3	1	2
4	1	2
5	0	3

Row 3 (i = 3)

j	smallest	point
1	0	3
2	1	2
3	2	1
4	1	2
5	0	3

as per above Row 4 & Row 5

Row 4 = Row 2 & Row 5 = Row

Q.6 Diamond with Continuous Numbers
for n = 4

```
        1
      2   3
    4   5   6
  7   8   9   10
    11  12  13
      14  15
        16
```

```
int n=4;
int count=1;
for (int i=1; i<=n; i++){
    for (int j=1; j<=n; j++){
        if (i+j>=n+1){
            sop(count+" ");
            count++;
        }else{
            sop(" ");
        }
    }
    sop();
}
for (int i=n-1; i>=1; i--){
    for (int j=1; j<=n; j++){
        if (i+j>=n+1){
            sop(count+" ")
        }else{
            sop();
        }
    }
    sop()
}
```

Row 1 (i=1)

j	i+j	condition	output	count
1	2	false	" "	1
2	3	false	" "	1
3	4	false	" "	1
4	5	True	"	2

Row 2 (i=2)

j	i+j	cond^n	o/p	count
1	3	false	" "	2
2	4	false	" "	2
3	5	True	2	3
4	6	True	3	4

Row 3 (i=3)

j	i+j	cond^n	o/p	count
1	4	false	" "	4
2	5	true	4	5
3	6	true	5	6
4	7	true	6	7

Row 4 (i=4)

j	i+j	cond^n	o/p	count
1	8	True	7	8
2	9	True	8	9
3	10	True	9	10
4	11	True	10	11